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Dear Editors of *Genes*,

Please consider our manuscript, "Cross-model uncertainty-aware minimal editing of cis-regulatory elements for cell-type-selective design," for publication as a Research Article. The study addresses a biological problem in functional genomics: how to retune natural cis-regulatory elements (CREs) with a small, interpretable number of nucleotide substitutions while preserving a cell-type-selective regulatory signal.

Existing sequence-design workflows have made sequence-to-activity prediction highly effective, but commonly select edits with the same predictor that defines the objective or return a single unconstrained high-scoring sequence. SafeEdit-CRE introduces a procedurally separated framework for minimal editing: candidate generation uses one predictor, cross-model review uses an architecture-diverse ensemble, and final evaluation uses a held-out multi-kernel model. It combines the published cell-type-specificity predictor with validation-calibrated uncertainty, sequence-domain controls, and constrained beam search on natural 200-nt CRE scaffolds, prioritizing variants with consistent predictions across separately trained, architecture-diverse models. The resulting compact panels are designed to reduce construct burden and focus reporter-screening resources on candidates that remain traceable to a natural regulatory scaffold.

The principal findings are:

- Across 600 held-out natural CRE parents in K562, HepG2, and SK-N-SH cells, SafeEdit-CRE increases cross-model specificity-margin gain from 0.822 for greedy editing to 0.877 (paired difference 0.055; 95% CI 0.040–0.071).
- Against a compute-matched primary-model beam, SafeEdit-CRE maintains closely matched predicted specificity (difference -0.004; 95% CI -0.011–0.004), while cross-model uncertainty decreases from 0.335 to 0.314.
- In a locked, non-overlapping 90-parent design-quality benchmark, the fraction passing all prespecified sequence and model-agreement checks increases from 38.8% for greedy editing to 60.6% for SafeEdit-CRE, yielding 74 Tier A variants prioritized for reporter screening.

These results make a focused contribution to the scope of *Genes*: they connect sequence modeling with cis-regulatory grammar, cell-type-specific transcriptional control, and reproducible functional-genomics design. They also provide a direct engineering output: a small, traceable set of low-edit CRE constructs that can be ordered and screened without rebuilding an unconstrained design space. The code, candidate library, and frozen summary inputs are assembled in the v1.1.0 release bundle and will be deposited on GitHub and Zenodo before publication. The complete bundle is available from the corresponding author during peer review; the final DOI and commit hash will be inserted in the published version.

The manuscript is original, is not under consideration elsewhere, and has been approved by the author. The author declares no competing interests. Thank you for considering this work.

Sincerely,